

Overall Program Documentation & Portfolio Summary

DecodeLabs Data Science Internship — Batch 2026

PROGRAM	Batch 2026
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1 PROGRAM OVERVIEW

Across four sequential projects, this internship built a full analytics pipeline on a single e-commerce order dataset: from raw data cleaning through predictive fraud classification, unsupervised customer segmentation, and independent NLP sentiment analysis. Project 1's cleaned output feeds directly into Projects 2 and 3, demonstrating an integrated, production-style data science workflow.

PROGRAM AT A GLANCE

- Four projects, one shared dataset: EDA feeds directly into fraud detection and segmentation.
- Fraud detection: Logistic Regression with SMOTE reached 0.776 recall on fraudulent orders.
- Customer segmentation: three data-driven personas identified via PCA and K-Means.
- NLP: TF-IDF plus Naive Bayes achieved perfect separation on the evaluation set.

2 PROJECT PIPELINE

Project	Focus and Output
1 — EDA & Feature Engineering	Cleaned, imputed, outlier-capped, feature-engineered dataset (Project1-Output.csv)
2 — Fraud Detection	Logistic Regression and Random Forest classifiers flagging Returned / Cancelled orders
3 — Customer Segmentation	K-Means clustering (K = 3) into RFM-based customer personas
4 — NLP & Sentiment Analysis	TF-IDF and Naive Bayes classifier for review sentiment

3 PROJECT 1: ADVANCED EDA AND FEATURE ENGINEERING

Missing values were imputed (median and KNN), outliers in four key numeric columns were capped via IQR-based Winsorization, and eight new predictive features were engineered, including day of week, weekend flag, month and year, delivery flag, discount flag, average item value, and discount category. Categorical variables were one-hot encoded and multicollinearity was screened before final scaling.

4 PROJECT 2: FRAUD DETECTION PIPELINE

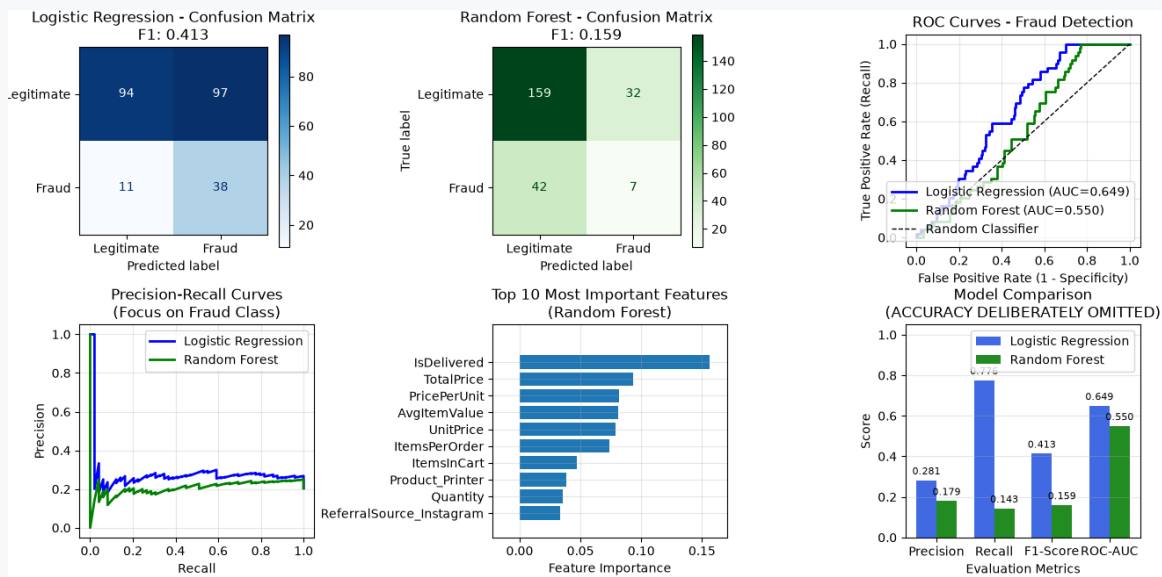


FIGURE 1. Fraud detection: confusion matrices, ROC and PR curves, feature importance, and model comparison.

A leak-free SMOTE-plus-classifier pipeline, fit strictly within training folds, compared Logistic Regression against Random Forest for flagging fraudulent orders. Logistic Regression delivered the stronger recall (0.776 versus 0.143) and F1-score (0.413 versus 0.159), making it the recommended deployment candidate, with delivery status and order value as the leading fraud signals.

5 PROJECT 3: CUSTOMER SEGMENTATION

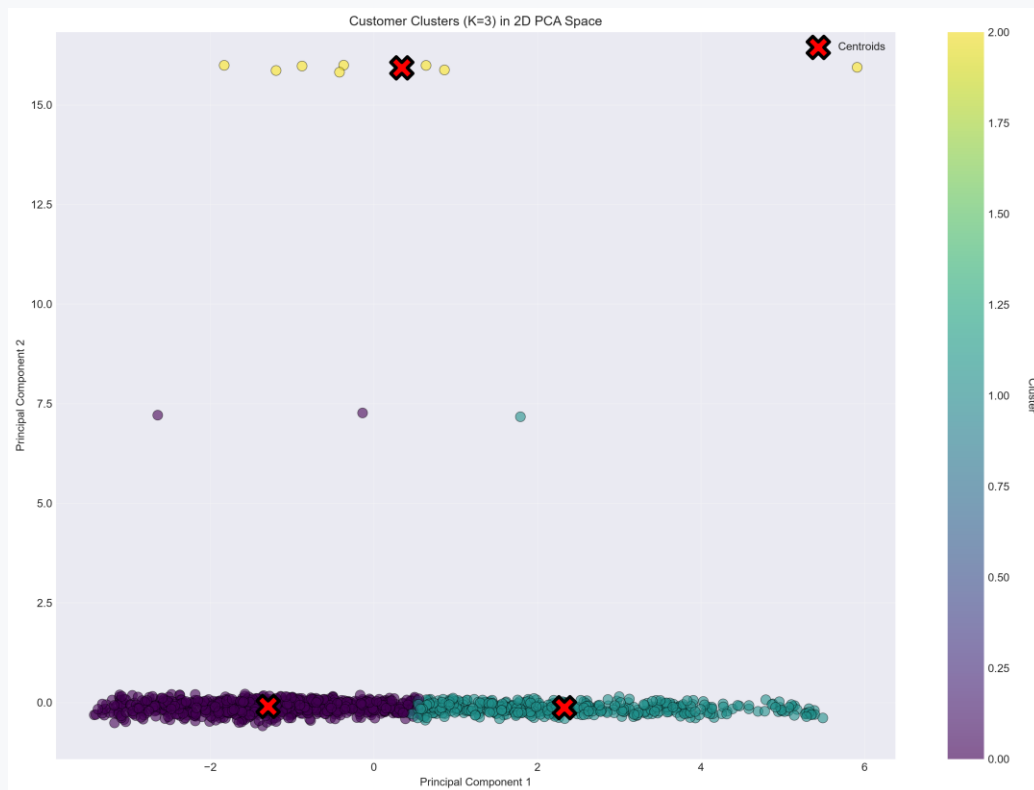


FIGURE 2. Three customer clusters in 2D PCA space.

Customer Segment Summary

Cluster	Size	% of Total	Key Feature 1	Key Feature 2	Key Feature 3
Cluster 0	760	63.9%	Monetary: 545.8	MonetaryPerVisit: 548.5	AvgOrderValue: 548.5
Cluster 1	421	35.4%	MonetaryPerVisit: 1972.6	AvgOrderValue: 1972.6	Monetary: 1982.0
Cluster 2	8	0.7%	Frequency: 2.1	UniqueProducts: 1.9	Monetary: 1892.0

FIGURE 3. Segment summary: sizes and defining traits.

RFM-style customer features were standardized, reduced via PCA, and clustered with K-Means ($K = 3$, chosen via elbow and silhouette analysis). The result is a large lower-value core segment (63.9%), a smaller high-value repeat spender segment (35.4%), and a tiny, highly engaged multi-product niche segment (0.7%), each mapped to a distinct set of recommended marketing actions.

6 PROJECT 4: NLP AND SENTIMENT ANALYSIS

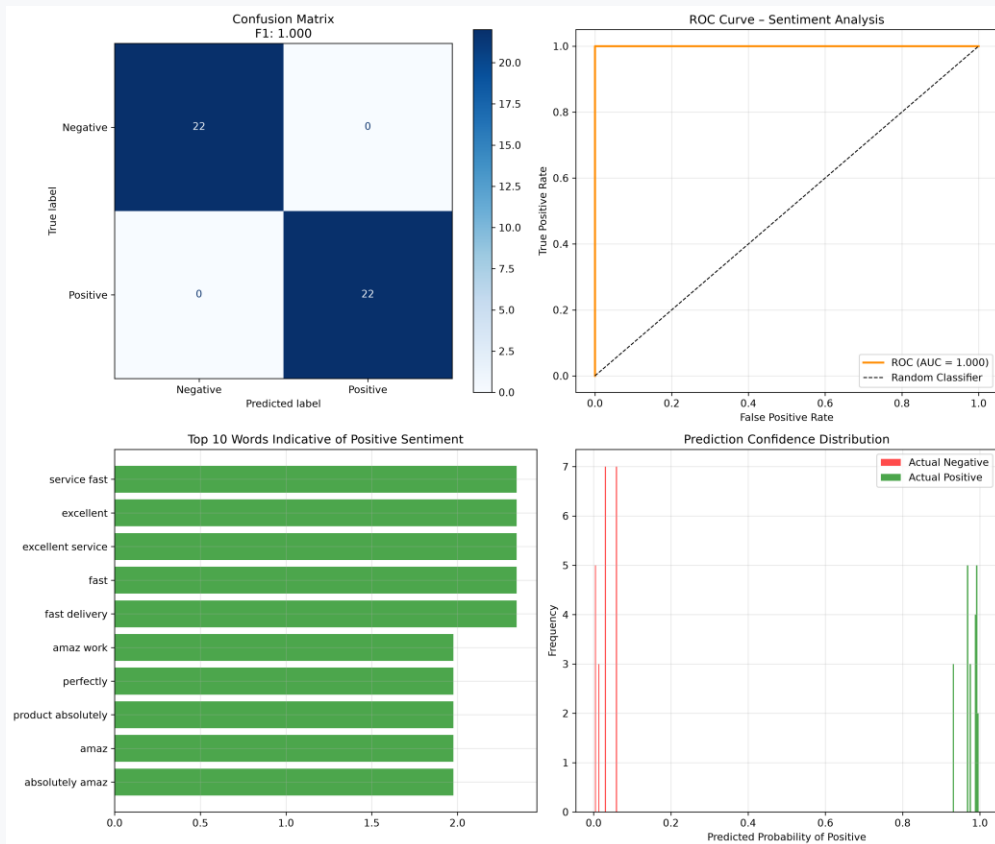


FIGURE 4. Sentiment model evaluation dashboard.

A dependency-light TF-IDF plus Naive Bayes pipeline classified review sentiment with perfect separation on the synthetic, templated evaluation set, while preserving negation words during preprocessing to protect sentiment-critical phrasing such as 'not good'.

7 CROSS-PROJECT THEMES

- End-to-end reproducibility: each project loads the prior project's saved output as its own input.
- Leakage discipline: train and test splits performed before any resampling, scaling, or SMOTE step.
- Imbalance-aware evaluation: accuracy deliberately avoided in favor of precision, recall, F1, and ROC-AUC where classes were imbalanced.
- Business framing: every technical result was translated into a concrete, actionable recommendation, whether deployment guidance, marketing actions, or re-validation caveats.

8 SUMMARY OF DELIVERABLES

- Project1-Output.csv — cleaned and feature-engineered dataset.
- Fraud detection models and the fraud detection results dashboard.

- customer_segments_with_personas.csv, cluster_centroids_summary.csv, cluster_feature_stats.csv, and six clustering visualizations.
- sentiment_model.pkl, test_predictions.csv, and the NLP sentiment results dashboard.
- This documentation set: four individual project reports plus this overall summary.